



THE KING'S SCHOOL

2005
Higher School Certificate
Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value



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2005 Higher School Certificate Trial Examination

Mathematics

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	(a), (c), (d), (e), (f)			(b)			12
2			(a)		(b)		12
3	(d)	(a)				(b), (c)	12
4				(a)		(b), (c)	12
5	(b)(ii)(iii)(iv)		(a)		(b)(i)		12
6	(b)		(a)(i)(ii)		(a)(iii)		12
7		(b)			(a)		12
8			(a)(i)		(b)ii	(a)(ii), (b)(i)(iii)	12
9	(a)		(b)(iv)(v)	(b)(ii)(iii)	(b)(i)		12
10			(a), (b)(iii)	(b)(i)(ii)	(b)(iv)(v)		12
Marks	27	10	26	15	25	17	120

Total marks – 120
Attempt Questions 1-10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

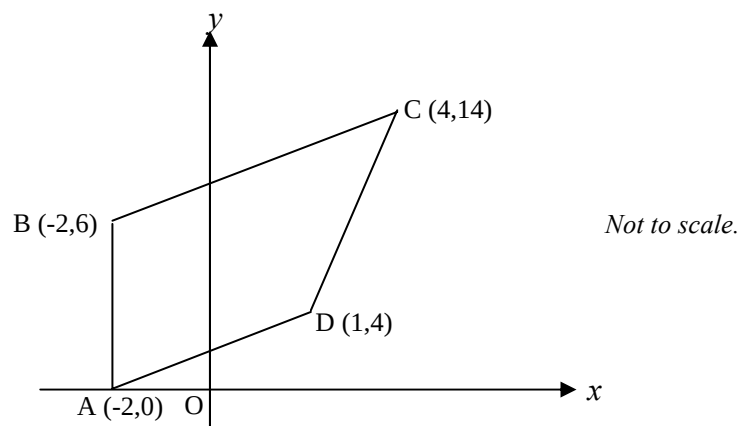
Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- (a) Convert 2005 degrees to radians, correct to the nearest integer. **1**
- (b) State the amplitude and period of the function $y = -\sin 2x$ **2**
- (c) Simplify $\frac{2}{3} - \frac{3x+8}{12}$ **2**
- (d) Express $\frac{\sqrt{2}+2}{\sqrt{2}}$ in simplest form. **2**
- (e) Find $\sum_{n=1}^3 \frac{6}{n}$ **1**
- (f) Solve
- (i) $|x-3| < 9$ **2**
- (ii) $2005^x = 21$, correct to 1 decimal place. **2**

End of Question 1

(a)



ABCD is a quadrilateral with vertices A(-2,0), B(-2,6), C(4,14) and D(1,4).

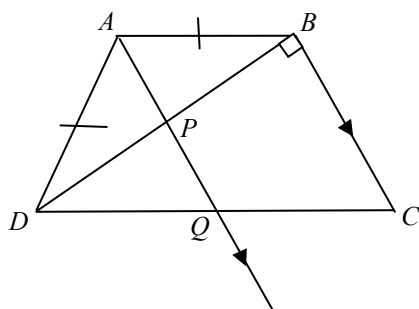
- | | | |
|-------|---|---|
| (i) | Show that AD is parallel to BC. | 2 |
| (ii) | Find the sum of the lengths of AD and BC. | 2 |
| (iii) | Show that the equation of the line AD is $3y - 4x - 8 = 0$ | 2 |
| (iv) | Hence, or otherwise, find the area of the quadrilateral ABCD. | 3 |

(b) Differentiate

- | | | |
|------|---------------------|---|
| (i) | $(2x + 1)^{-1}$ | 1 |
| (ii) | $\frac{x}{x^2 + 1}$ | 2 |

End of Question 2

(a)



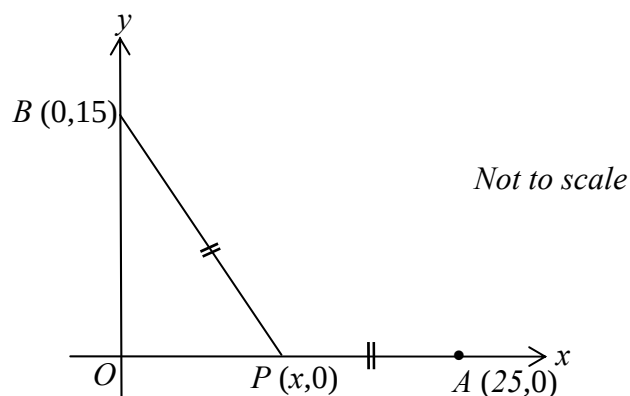
In the diagram, $AB = AD$ and $\angle DBC = 90^\circ$.

Line APQ is parallel to line BC and meets DB at P and DC at Q .

- | | |
|---|----------|
| (i) Give a reason why $\angle APB = 90^\circ$. | 1 |
| (ii) Prove that $\triangle ABP$ is congruent to $\triangle ADP$. | 2 |
| (iii) Deduce that $DQ = QC$. | 2 |
-
- | | |
|--|----------|
| (b) Find a primitive function of $\frac{x^3}{x^4 + 1}$ | 1 |
|--|----------|
-
- | | |
|---|----------|
| (c) Evaluate $\int_0^{\frac{\pi}{8}} 2 \sec^2 2x \, dx$ | 2 |
|---|----------|

Question 3 continues on next page

(d)



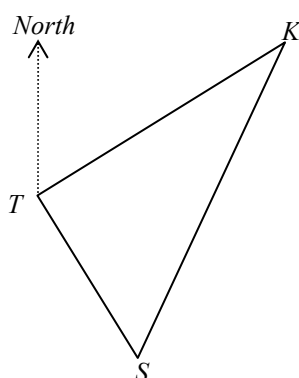
$P(x,0)$ is the point between $O(0,0)$ and $A(25,0)$ such that $PA = PB$, where $B = (0,15)$.

(i) Explain why $PB = 25 - x$. 1

(ii) Find the value of x . 3

End of Question 3

(a)



The bearing of K from T is 070° and $TK = 400\text{km}$.

The bearing of S from T is 150° and $TS = 213\text{km}$.

- | | |
|--|----------|
| (i) Show that $\angle KTS = 80^\circ$. | 1 |
| (ii) Find KS , nearest km. | 2 |
| (iii) Use the sine rule to find the bearing of S from K , nearest degree. | 3 |
| (b) Use Simpson's Rule once to approximate $\int_1^2 x \sin x \, dx$,
correct to 2 decimal places. | |
| 3 | |
| (c) The curve $y = f(x)$ has a stationary point at $x = 1$ and $f''(x) = 6x - 2$. | |
| Find the x coordinate at which there is another stationary point. | |
| 3 | |

End of Question 4

(a) The quadratic equation $Q(x) = 2x^2 + (k - 3)x + (k + 3) = 0$ has no real roots.

(i) Prove that $k^2 - 14k - 15 < 0$ 2

(ii) Find the values of k . 2

(iii) For what values of k is $Q(x) > 0$ for all real values of x ? 1

(b) The population P of a country town on January 1, 1995 was 1730 and on January 1, 2005 was 1160. The town's population is known to be changing according to the equation $\frac{dP}{dt} = kP$ where t is time in years measured from January 1, 1995 and k is a constant.

(i) Verify that $P = Ae^{kt}$, where A is a constant, satisfies the equation $\frac{dP}{dt} = kP$. 2

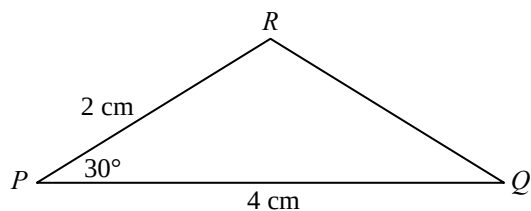
(ii) State the value of A 1

(iii) Find the value of k , correct to 1 significant figure. 2

(iv) Approximately how many people are expected to leave the town during the year 2005? 2

End of Question 5

(a)



$\triangle PQR$ has a fixed angle $P = 30^\circ$ but side PR is increasing at 4cm/min and side PQ at 2cm/min .

Initially, $PR = 2\text{cm}$ and $PQ = 4\text{cm}$.

Let $A(t)$ be the area of $\triangle PQR$ after t minutes.

(i) Show that $A(t) = 2t^2 + 5t + 2$ 2

(ii) When is the area of $\triangle PQR$ equal to 102cm^2 ? 2

(iii) At what rate is the area of $\triangle PQR$ changing after 10 minutes? 2

Question 6 continues next page

- (b) George borrows \$20 000 from Saint Bank. This loan plus interest is to be repaid in equal monthly instalments of \$399 over five years. Interest of 7.2% p.a. is compounded monthly on the balance owing at the start of each month.

Let $\$A_n$ be the amount owing after n months.

- (i) Over the five year repayment period, how much interest is charged? **1**
- (ii) Show that $A_1 = 19721$. **1**
- (iii) Clearly show that $A_2 = 20\,000 \times 1.006^2 - 399(1 + 1.006)$. **1**
- (iv) Deduce that $A_n = 66\,500 - 46\,500 \times 1.006^n$. **2**
- (v) After two years of repayments George decides on the very next day to repay the loan in full by one payment. **1**
- How much will this one payment be?

End of Question 6

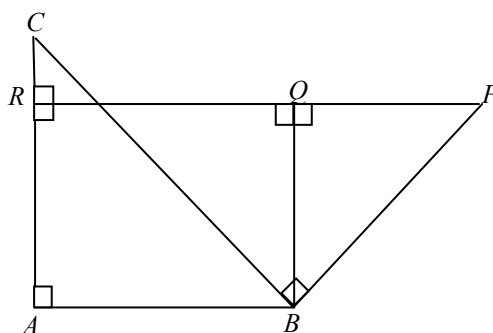
(a) Consider the function $f(x) = 8x^3 - 3x^4$.

(i) For what values of x is the function decreasing? 2

(ii) Show that the point $(0,0)$ is a point of inflection. 2

(iii) Sketch the function showing x intercepts and stationary points. 3

(b)



In the diagram there are six right angles marked.

(i) Prove that $\triangle ABC$ is similar to $\triangle BPQ$. 2

(ii) If further, $BC = BP$, which test proves that $\triangle ABC$ is congruent to $\triangle BPQ$? 1

(iii) Deduce that $PR = AB + AC$. 2

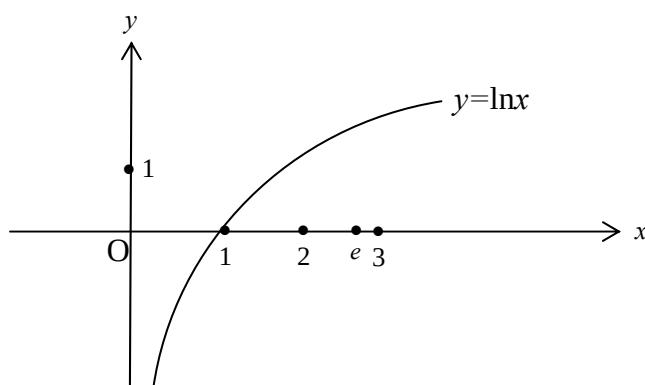
End of Question 7

- (a) The line $y = 3 - 2x$ and the parabola $y = 2x - x^2$ meet at the points (1,1) and (3,-3).

[Do Not Show This]

- (i) On the same diagram sketch the line and the parabola and include on your diagram the x intercepts for each. 2
- (ii) Find the area of the region bounded by the line and the parabola. 3

(b)



- (i) The region bounded by the curve $y = \ln x$ and the y axis between $y = 0$ and $y = 1$ is revolved about the y axis.
Find the volume of the solid of revolution. 3
- (ii) Show that $\frac{d}{dx} \left[x \left((\ln x)^2 - 2 \ln x + 2 \right) \right] = (\ln x)^2$ 2
- (iii) The region bounded by the curve $y = \ln x$ and the x axis between $x = 1$ and $x = e$ is revolved about the x axis.
Find the volume of the solid of revolution. 2

End of Question 8

- (a) The sum of the first n terms of an arithmetic series is given by $S_n = n(n + a)$, where a is a constant.

Find the common difference.

2

- (b) A particle is moving along the x axis. Initially it is at the origin with velocity 1.

Its displacement at any time $t \geq 0$ is given by $x = \sin^2 t + \sin t$.

- (i) Show that its velocity at any time $t \geq 0$ is given by $\dot{x} = \cos t(2 \sin t + 1)$

2

- (ii) Find the first five times that the particle stops.

3

- (iii) The acceleration of the particle can be expressed as $\ddot{x} = 2 \cos 2t - \sin t$

[DO NOT SHOW THIS]

Find the displacement, velocity and acceleration when $t = \frac{\pi}{2}$

2

- (iv) Use (iii) to explain why immediately after $t = \frac{\pi}{2}$ the particle will move towards the origin.

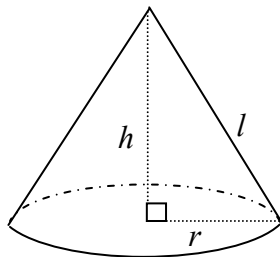
1

- (v) Find the least distance that the particle moves between its stopping positions.

2

End of Question 9

(a)



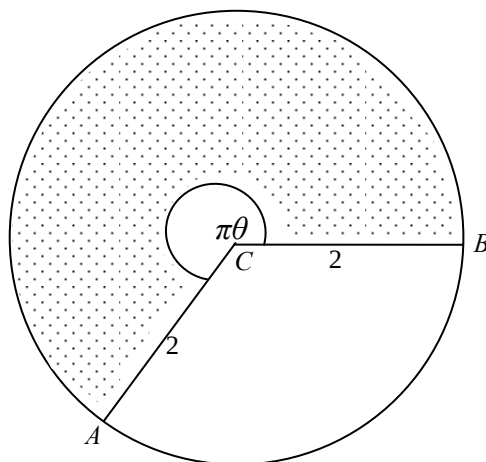
A cone has radius r , height h and slant height l .

The volume of the cone $V = \frac{\pi}{3} r^2 h$

Show that the volume of the cone can be expressed as $V = \frac{\pi}{3} \sqrt{l^2 r^4 - r^6}$.

2

(b)



The angle at the centre C of a circle of radius 2cm is $\pi\theta$ radians, $0 < \theta < 2$, as shown in the diagram.

- (i) Write down the length of the arc AB of this sector. 1
- (ii) This sector is cut from the circle along the radii CA and CB and folded to make a cone. 1
- Find the radius of the cone. 1
- (iii) Show that the volume of the cone is given by $V = \frac{\pi}{3} \sqrt{4\theta^4 - \theta^6}$ 1
- (iv) Find the value of θ , correct to two decimal places, for which the volume of the cone is a maximum. 4
- (v) Sketch the graph of V against θ using the same scale on both axes. 3

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Qn 1

$$(a) \quad 2005 \times \frac{\pi}{180} \text{ radians} = 35^\circ$$

$$(b) \quad \text{amplitude} = 1, \quad \text{period} = \frac{2\pi}{2} = \pi$$

$$(c) \quad \frac{8 - (3x + 8)}{12} = -\frac{3x}{12} = -\frac{x}{4}$$

$$(d) \quad \frac{\sqrt{2} + 2}{\sqrt{2}} = 1 + \frac{2}{\sqrt{2}} = 1 + \sqrt{2}$$

$$(e) \quad 6 + 3 + 2 = 11$$

$$(f) \quad (i) \quad -9 < x - 3 < 9 \\ \therefore -6 < x < 12$$

$$(ii) \quad x \ln 2005 = \ln 21$$

$$x = \frac{\ln 21}{\ln 2005} = 0.4, \quad 1 \text{ d.p.}$$

Ques 2

(a) (i) gradient $AD = \frac{4}{1-2} = \frac{4}{3}$

$$\text{grad } BC = \frac{14-6}{4-2} = \frac{8}{2} = \frac{4}{1}$$

$\therefore AD \parallel BC$ (equal grads)

(ii) $AD + BC = \sqrt{3^2 + 4^2} + \sqrt{6^2 + 8^2}$
 $= 5 + 10 = 15$

(iii) AD is $y = \frac{4}{3}(x-2)$

$$\text{or } 3y = 4x - 8$$

$$\text{i.e. } 3y - 4x + 8 = 0$$

(iv) $ABCD$ is a trapezium, from (i)

$\perp$ distance from $B(-2, 6)$ to AD

$$= \frac{18 + 8 - 8}{\sqrt{3^2 + 4^2}} = \frac{18}{5}$$

$$\therefore \text{Area} = \frac{1}{2} (15) \frac{18}{5} = 27 \text{ u}^2$$

(b) (i) $-(2x+1)^{-2} \cdot 2 = \frac{-2}{(2x+1)^2}$

(ii) $\frac{(x^2+1) \cdot 1 - x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$

Ques 3

(a) (i) $\angle APB = \angle PBC$, alt $\angle$ s in $\parallel$ lines
 $= 90^\circ$, given

(ii) Since $\angle APB = 90^\circ$ then $\angle APD = 90^\circ$, $\angle DPB$ straight

$AD = AB$ (Hypotenuses), given
 AP is common

$\therefore \Delta$ s congruent, RHS

(iii) $DP = PB$, corr. sides in cong Δ s (ii)

$\therefore DQ = QC$, ratio intercept theorem in $\parallel$ lines

(b) $\frac{1}{4} \int \frac{4x^3}{x^4+1} dx = \frac{1}{4} \ln(x^4+1)$

(c) $\int_0^{\frac{\pi}{8}} 2 \sec^2 2x \, dx = \frac{2}{2} [\tan 2x]_0^{\frac{\pi}{8}}$
 $= \tan \frac{\pi}{4} - 0$
 $= 1$

(d) (i) $PA = 25 - x \quad \therefore PB = 25 - x$ since $PA = PB$

(ii) In ΔOPB , $x^2 + 15^2 = (25 - x)^2$

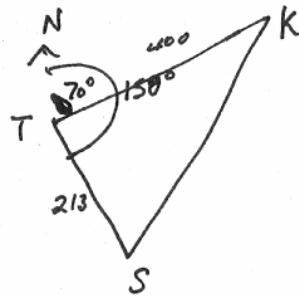
ie. $x^2 + 225 = 625 - 50x + x^2$

or $50x = 400$

$x = 8$

Q 4

(a) (i)



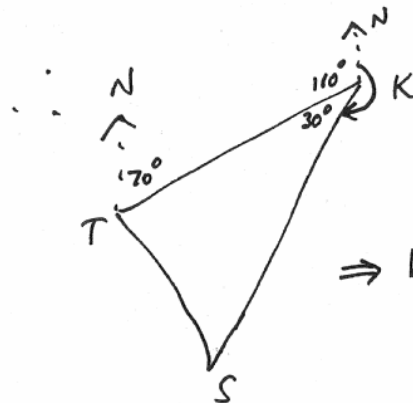
$$\therefore \angle KTS = 150^\circ - 70^\circ = 80^\circ$$

$$(ii) \quad KS^2 = 400^2 + 213^2 - 2 \times 400 \times 213 \cos 80^\circ$$

$$\Rightarrow KS = 419 \text{ km}$$

$$(iii) \quad \frac{\sin K}{213} = \frac{\sin 80^\circ}{419}$$

$$\therefore \sin K = \frac{213 \sin 80^\circ}{419} \Rightarrow \hat{K} = 30^\circ, \text{ nearest degree}$$



$$\Rightarrow \text{bearing of S from K} = 360^\circ - 140^\circ = 220^\circ$$

$$(b) \quad \int_1^2 x \sin x \, dx \approx \frac{1}{6} \cdot 1 \left[\sin 1 + 2 \sin 2 + 4 \cdot \frac{3}{2} \sin 1.5 \right]$$

$$\approx 1.44$$

$$(c) \quad \therefore f'(x) = 3x^2 - 2x + c \quad : \quad f'(1) = 3 - 2 + c = 0$$

$$c = -1$$

$$\therefore f'(x) = 3x^2 - 2x - 1 = (x-1)(3x+1)$$

$$\Rightarrow \text{stat. pt when } 3x+1=0 \text{ i.e. } x = -\frac{1}{3}$$

Qn 5

(a) (i) We must have $\Delta < 0$

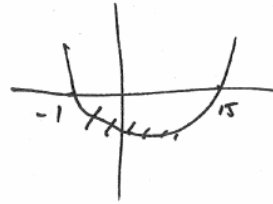
$$\therefore (k-3)^2 - 4 \times 2(k+3) < 0$$

$$\text{i.e. } k^2 - 6k + 9 - 8k - 24 < 0$$

$$\text{or } k^2 - 14k - 15 < 0$$

(ii) From (i), $(k+1)(k-15) < 0$

$$\Rightarrow -1 < k < 15$$



(iii) i.e. positive definite $\Rightarrow \Delta < 0$

$$\therefore -1 < k < 15$$

(b) (i) If $P = Ae^{kt}$ then $\frac{dP}{dt} = Ake^{kt}$
 $= k(Ae^{kt})$
 $= kP$

(ii) $t=0, P=1730 \Rightarrow A=1730$

(iii) $t=10, P=1160 \Rightarrow 1160 = 1730 e^{10k}$

$$\therefore e^{10k} = \frac{116}{173} \Rightarrow 10k = \ln\left(\frac{116}{173}\right)$$

$$\text{i.e. } k = \frac{1}{10} \ln\left(\frac{116}{173}\right) = -0.04, \text{ 1 sig. fig.}$$

(iv) Jan, 2006 $\Rightarrow t=11 \therefore P = 1730 e^{-0.04 \times 11} = 1730 e^{-0.44}$

$$\approx 1114 \quad (1110 \text{ will do})$$

$$\therefore \text{expect } 1160 - 1114 = 46 \text{ to leave } (50 \text{ will do})$$

Ques 6

(a) (i) After t minutes, $PR = 2 + 4t$ and $PQ = 4 + 2t$

$$\begin{aligned}\therefore A(t) &= \frac{1}{2} (2+4t)(4+2t) \sin 30^\circ \\ &= \frac{1}{4} (2+4t)(4+2t) \\ &= (1+2t)(2+t) \\ &= 2t^2 + 5t + 2\end{aligned}$$

(ii) $2t^2 + 5t + 2 = 102$

$$\Rightarrow 2t^2 + 5t - 100 = 0$$

$$\begin{aligned}\therefore t &= \frac{-5 + \sqrt{25 + 800}}{4} \quad \text{since } t > 0 \\ &= \frac{-5 + \sqrt{825}}{4} \text{ min} \quad [\approx 5.93 \text{ min}]\end{aligned}$$

(iii) $\frac{dA}{dt} = 4t + 5 = 4 \times 10 + 5$ after 10 minutes
 $= 45 \text{ cm}^2/\text{min}$

(b) (i) Interest $= \$399 \times 12 \times 5 - \$20000 = \$3940$

(ii) $A_1 = 20000 \times 1.006 - 399 = 19721$
 $[* 7.2\% \text{ p.a.} = \frac{0.072}{12} \text{ p. month} = 0.006]$

(iii) $A_2 = A_1(1.006) - 399$
 $= (20000 \times 1.006 - 399)1.006 - 399$
 $= 20000 \times 1.006^2 - 399(1 + 1.006)$

(iv) (iii) $\Rightarrow A_n = 20000 \times 1.006^n - 399(1 + 1.006 + \dots + 1.006^{n-1})$
 $= 20000 \times 1.006^n - 399 \left(\frac{1.006^n - 1}{0.006} \right)$
 $= 20000 \times 1.006^n - 66500(1.006^n - 1)$
 $= 66500 - 46500 \times 1.006^n$

(v) The one payment $= A_{24} = 66500 - 46500 \times 1.006^{24} \approx \12821
[or $\$12820.99$]

Ques 7

$$(a) (i) f'(x) = 24x^2 - 12x^3 \\ = 12x^2(2-x) < 0 \quad \text{if } 2-x < 0 \\ \text{ie } x > 2$$

$\therefore$ decreasing for $x > 2$ [$x \geq 2$ will do]

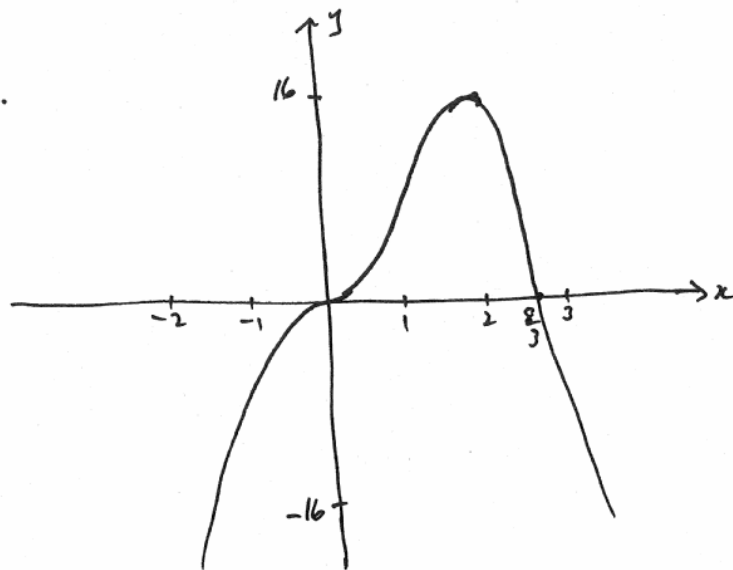
$$(ii) f''(x) = 48x - 36x^2 = 12x(4-3x)$$

Now, $f''(0) = 0$ and $f''(-1) < 0$, $f''(1) > 0$ [change in concavity]

$\therefore (0,0)$ is a pt of inflection

$$(iii) f'(x) = 0 \Rightarrow x = 0, 2 \quad \text{ie start pts } (0,0), (2,16) \\ y = 0, 16$$

$$f(x) = 0 \Rightarrow x^3(8-3x) = 0 \Rightarrow x \text{ intercepts } 0, \frac{8}{3}$$



(b) (i) (Lots of alternatives)

$$\angle ABQ = 90^\circ, \text{ angle sum of } \triangle ABQR$$

$$\therefore \angle ABC = \angle PBQ, \text{ both complements of } \angle CBQ$$

$$\text{and } \angle A = \angle Q = 90^\circ, \text{ given}$$

$$\therefore \triangle ABC \parallel \triangle BPQ, \text{ 2 angles equal}$$

(ii) AAS

$$(iii) PR = PQ + QR$$

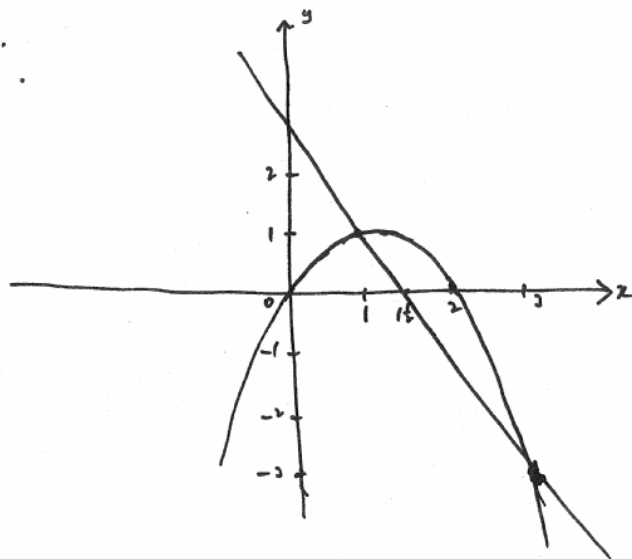
$$= AC + QR, \text{ cong } \triangle s \text{ in (ii)}$$

$$= AC + AB, \text{ opp sides of rectangle } ABQR$$

Que 8

(a) (i) For $y = 3 - 2x$, $y = 0 \Rightarrow x = \frac{3}{2}$

For $y = 2x - x^2 = x(2 - x)$, $y = 0 \Rightarrow x = 0, 2$



$$(ii) A = \int_1^3 (2x - x^2 - (3 - 2x)) dx$$

$$= \int_1^3 (4x - x^2 - 3) dx$$

$$= \left[2x^2 - \frac{x^3}{3} - 3x \right]_1^3$$

$$= 18 - 9 - 9 - \left(2 - \frac{1}{3} - 3 \right) = \frac{4}{3} \text{ unit}^2$$

$$(b) (i) V = \pi \int_0^1 x^2 dy \quad : \quad y = \log_e x \Rightarrow x = e^y \\ \therefore x^2 = e^{2y}$$

$$\therefore V = \pi \int_0^1 e^{2y} dy = \frac{\pi}{2} [e^{2y}]_0^1 \\ = \frac{\pi}{2} (e^2 - 1) \quad u^3$$

$$(ii) \frac{d}{dx} x ((\ln x)^2 - 2 \ln x + 2) = x \left(2 \ln x \cdot \frac{1}{x} - \frac{2}{x} \right) + ((\ln x)^2 - 2 \ln x + 2) \cdot 1 \\ = 2 \ln x - 2 + (\ln x)^2 - 2 \ln x + 2 \\ = (\ln x)^2$$

$$(iii) V = \pi \int_1^e y^2 dx = \pi \int_1^e (\ln x)^2 dx \\ = \pi \left[x ((\ln x)^2 - 2 \ln x + 2) \right]_1^e \quad \text{from (ii)} \\ = \pi (e(1 - 2 + 2) - 1(0 - 0 + 2)) \\ = \pi (e - 2) \quad u^3$$

Ques 9

$$(a) S_1 = T_1 = 1 + a$$

$$S_2 = 2(2+a) = 4 + 2a$$

$$\therefore T_2 = 4 + 2a - (1+a) = 3 + a$$

$$\therefore \text{common difference} = 3 + a - (1+a) = 2$$

$$(b) (i) \dot{x} = 2 \sin t \cos t + \cos t$$

$$\text{ie } \dot{x} = \cos t (2 \sin t + 1)$$

$$(ii) \dot{x} = 0 \Rightarrow \cos t = 0 \text{ or } \sin t = -\frac{1}{2}$$

$$\therefore t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\text{or } t = \frac{7\pi}{6}, \frac{11\pi}{6}, \dots$$



$$\text{ie. first 5 times are } \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}, \frac{5\pi}{2}$$

$$(iii) t = \frac{\pi}{2}, x = 1 + 1 = 2$$

$$\dot{x} = 0$$

$$\ddot{x} = 2 \cos \pi - 1 = -2 - 1 = -3$$

(iv) Since $\ddot{x} = -3 < 0$ then $\dot{x}$ will decrease
ie. $\dot{x}$ will become negative

$\Rightarrow$ particle moves from $x = 2$ towards the origin

$$(v) t = \frac{7\pi}{6}, x = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

$$t = \frac{3\pi}{2}, x = 1 - 1 = 0$$

$$t = \frac{11\pi}{6}, x = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

$$t = \frac{5\pi}{2}, x = 1 + 1 = 2$$

$$\Rightarrow \text{least distance} = \frac{1}{4}$$

$$(v) \quad V = \frac{\pi}{3} \sqrt{4\theta^4 - \theta^6}, \quad 0 < \theta < 2$$

$$\theta \rightarrow 0, V \rightarrow 0$$

$$\theta \rightarrow 2, V \rightarrow 0$$

$$\theta = 1.63, V = \frac{\pi}{3} \sqrt{4(1.63)^4 - 1.63^6} \approx 3.22$$

$$[\theta = 1, V \approx 1.8]$$

